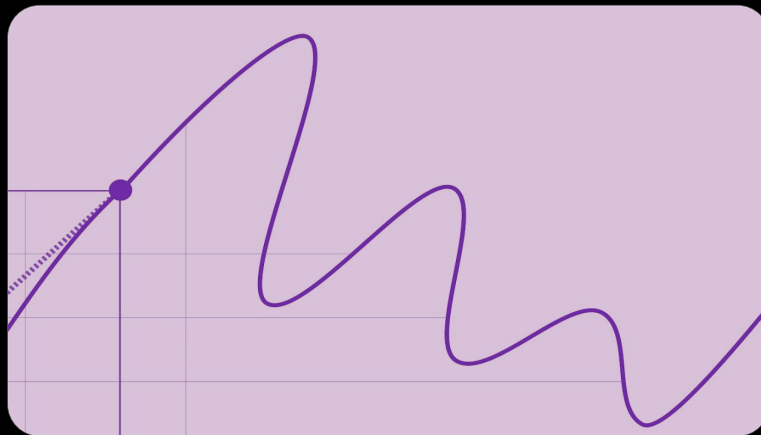
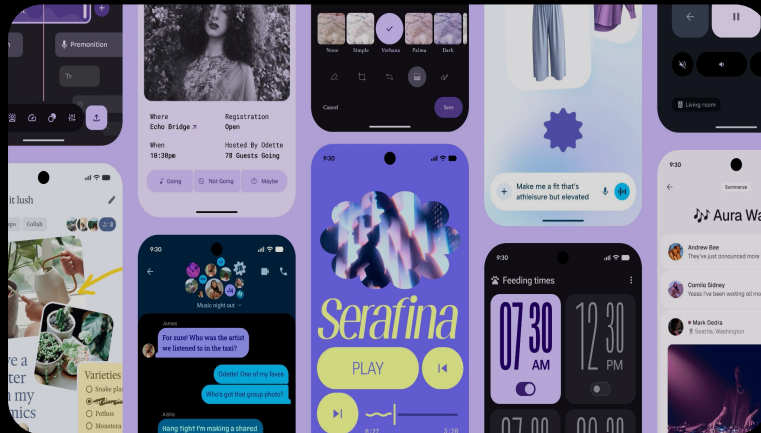


App Contador ViewModel

Prof. Dr. Fernando Sambinelli
IFSP Hortolândia

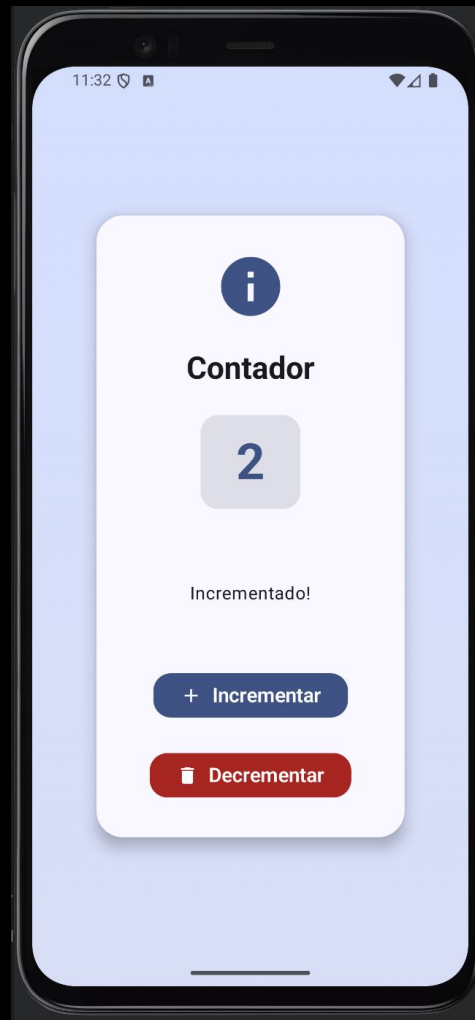
Vamos Começar



App Contador ViewModel

Praticar Uso de ViewModel

- × Ideia do App: o mesmo app implementado com estados ajustado para padrão ViewModel



Dependências

Estrutura de Dependências

1	[versions]
2	agp = "8.6.2"
3	kotlin = "2.0.21"
4	coreKtx = "1.15.0"
5	junit = "4.13.2"
6	junitVersion = "1.3.0"
7	espressoCore = "3.7.0"
8	lifecycleRuntimeKtx = "2.8.6"
9	activityCompose = "1.9.3"
10	composeBom = "2024.10.01"
11	
12	lifecycleViewmodelCompose = "2.8.6"
13	lifecycleRuntimeCompose = "2.8.6"
14	
15	[libraries]
16	androidx-core-ktx = { group = "androidx.core", name = "core-ktx", version.ref = "coreKtx" }
17	junit = { group = "junit", name = "junit", version.ref = "junit" }
18	androidx-junit = { group = "androidx.test.ext", name = "junit", version.ref = "junitVersion" }
19	androidx-espresso-core = { group = "androidx.test.espresso", name = "espresso-core", version.ref = "espressoCore" }
20	androidx-lifecycle-runtime-ktx = { group = "androidx.lifecycle", name = "lifecycle-runtime-ktx", version.ref = "lifecycleRuntimeKtx" }
21	androidx-activity-compose = { group = "androidx.activity", name = "activity-compose", version.ref = "activityCompose" }
22	androidx-compose-bom = { group = "androidx.compose", name = "compose-bom", version.ref = "composeBom" }
23	androidx-ui = { group = "androidx.compose.ui", name = "ui" }
24	androidx-ui-graphics = { group = "androidx.compose.ui", name = "ui-graphics" }
25	androidx-ui-tooling = { group = "androidx.compose.ui", name = "ui-tooling" }
26	androidx-ui-tooling-preview = { group = "androidx.compose.ui", name = "ui-tooling-preview" }
27	androidx-ui-test-manifest = { group = "androidx.compose.ui", name = "ui-test-manifest" }
28	androidx-ui-test-junit4 = { group = "androidx.compose.ui", name = "ui-test-junit4" }
29	androidx-material3 = { group = "androidx.compose.material3", name = "material3" }
30	
31	androidx-lifecycle-viewmodel-compose = { group = "androidx.lifecycle", name = "lifecycle-viewmodel-compose", version.ref = "lifecycleViewmodelCompose" }
32	androidx-lifecycle-runtime-compose = { group = "androidx.lifecycle", name = "lifecycle-runtime-compose", version.ref = "lifecycleRuntimeCompose" }
33	
34	[plugins]
35	android-application = { id = "com.android.application", version.ref = "agp" }
36	kotlin-android = { id = "org.jetbrains.kotlin.android", version.ref = "kotlin" }
37	kotlin-compose = { id = "org.jetbrains.kotlin.plugin.compose", version.ref = "kotlin" }

```
build.gradle.kts (app) ×

You can use the Project Structure dialog to view and edit your project configuration

7      android {
21          buildTypes {
22              release {
23                  isMinifyEnabled = false
24                  proguardFiles(
25                      ..files = getDefaultProguardFile( name = "proguard-android-optimize.txt"),
26                      "proguard-rules.pro"
27                  )
28              }
29          }
30          compileOptions {
31              sourceCompatibility = JavaVersion.VERSION_11
32              targetCompatibility = JavaVersion.VERSION_11
33          }
34          kotlinOptions {
35              jvmTarget = "11"
36          }
37          buildFeatures {
38              compose = true
39          }
40      }
41
42      dependencies {
43
44          implementation(libs.androidx.core.ktx)
45          implementation(libs.androidx.lifecycle.runtime.ktx)
46          implementation(libs.androidx.activity.compose)
47          implementation(platform( dependencyProvider = libs.androidx.compose.bom))
48          implementation(libs.androidx.ui)
49          implementation(libs.androidx.ui.graphics)
50          implementation(libs.androidx.ui.tooling.preview)
51          implementation(libs.androidx.material3)
52          testImplementation(libs.junit)
53          androidTestImplementation(libs.androidx.junit)
54          androidTestImplementation(libs.androidx.espresso.core)
55          androidTestImplementation(platform( dependencyProvider = libs.androidx.compose.bom))
56          androidTestImplementation(libs.androidx.ui.test.junit4)
57          debugImplementation(libs.androidx.ui.tooling)
58          debugImplementation(libs.androidx.ui.test.manifest)
59
60          // Lifecycle & ViewModel
61          implementation(libs.androidx.lifecycle.viewmodel.compose)
62          implementation(libs.androidx.lifecycle.runtime.compose)
63      }
```

Contador ViewModel

Estrutura do ViewModel

Implementações

Vamos criar uma classe de dados e o
ViewModel para o Contador

```
8      data class ContadorUiState( 3 Usages
9          val contador: Int = 0,
10         val message: String = "Clique nos botões para contar"
11     )
12
13     class ContadorViewModel : ViewModel() { 1 Usage
14         private val _uiState = MutableStateFlow( value = ContadorUiState()) 6 Usages
15         val uiState: StateFlow<ContadorUiState> = _uiState.asStateFlow() 1 Usage
16
17         fun incrementar() { 1 Usage
18             val estadoAtual = _uiState.value
19             _uiState.value = estadoAtual.copy(
20                 contador = estadoAtual.contador + 1,
21                 message = "Incrementado!"
22             )
23         }
24
25         fun decrementar() { 1 Usage
26             val estadoAtual = _uiState.value
27             if (estadoAtual.contador > 0) {
28                 _uiState.value = estadoAtual.copy(
29                     contador = estadoAtual.contador - 1,
30                     message = "Decrementado!"
31                 )
32             }
33         }
34
35         fun resetar() {
36             _uiState.value = ContadorUiState()
37         }
38     }
```

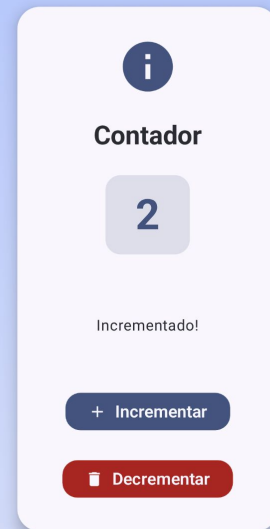
Função Principal

Estrutura Base do Layout

Implementações

A função principal não tem alterações.

```
43 class MainActivity : ComponentActivity() { 1 Usage
44     override fun onCreate(savedInstanceState: Bundle?) {
45         super.onCreate(savedInstanceState)
46         enableEdgeToEdge()
47         setContent {
48             AppContadorTheme {
49                 ContadorApp()
50             }
51         }
52     }
53 }
54
55 @Composable 2 Usages
56 //fun ContadorApp() {
57     //var contador by remember { mutableStateOf(0) }
58 fun ContadorApp(
59     viewModel: ContadorViewModel = viewModel()
60 ) {
61     val uiState by viewModel.uiState.collectAsStateWithLifecycle()
62
63     Box(
64         modifier = Modifier
65             .fillMaxSize()
66             .background(
67                 Brush.verticalGradient(
68                     colors = listOf(
69                         MaterialTheme.colorScheme.primaryContainer,
70                         MaterialTheme.colorScheme.secondaryContainer
71                     )
72                 )
73             ),
74         contentAlignment = Alignment.Center
75     ) {
76         Card(
77             modifier = Modifier.padding( all = 32.dp ),
78             elevation = CardDefaults.cardElevation( defaultElevation = 8.dp ),
79             shape = RoundedCornerShape( size = 24.dp ),
80             colors = CardDefaults.cardColors(
81                 containerColor = MaterialTheme.colorScheme.surface
82             )
83         ) {
84             Column(
85                 modifier = Modifier.padding( all = 32.dp ),
86                 horizontalAlignment = Alignment.CenterHorizontally,
87                 verticalArrangement = Arrangement.spacedBy( space = 24.dp )
88             ) {
```



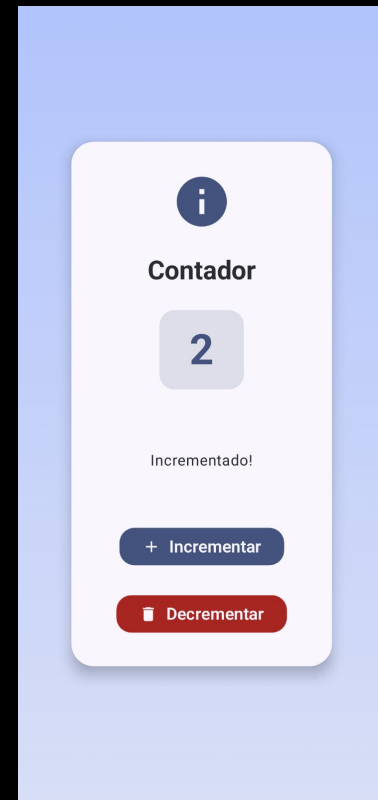
Outros Elementos

Estrutura do Layout

Implementações

Agora, fizemos os ajustes para usar o ViewModel. Também adicionamos um text de mensagem.

```
88
89
90 // ícone decorativo
91 Icon(
92   imageVector = Icons.Default.Info,
93   contentDescription = "Ícone de Interrogação",
94   modifier = Modifier.size( size = 64.dp),
95   tint = MaterialTheme.colorScheme.primary
96 )
97
98 Text(
99   text = "Contador",
100   style = MaterialTheme.typography.headLineMedium,
101   color = MaterialTheme.colorScheme.onSurface,
102   fontWeight = FontWeight.Bold
103 )
104
105 // Card para o número
106 Card(
107   shape = RoundedCornerShape( size = 16.dp),
108   colors = CardDefaults.cardColors(
109     containerColor = MaterialTheme.colorScheme.surfaceVariant
110   )
111 ) {
112   Text(
113     //text = "$contador",
114     text = "${uiState.contador}",
115     modifier = Modifier.padding(horizontal = 32.dp, vertical = 16.dp),
116     style = MaterialTheme.typography.displayMedium,
117     color = MaterialTheme.colorScheme.primary,
118     fontWeight = FontWeight.Bold
119   )
120
121   Spacer(modifier = Modifier.height( height = 16.dp))
122
123   Text(
124     text = uiState.message,
125     fontSize = 16.sp
126   )
127
128   Spacer(modifier = Modifier.height( height = 8.dp))
129
130 // Botão Incrementar
131 Button(
132   // onClick = { contador++ },
133   onClick = { viewModel.incrementar() },
134   colors = ButtonDefaults.buttonColors(
135     containerColor = MaterialTheme.colorScheme.primary
136   ),
137 )
```



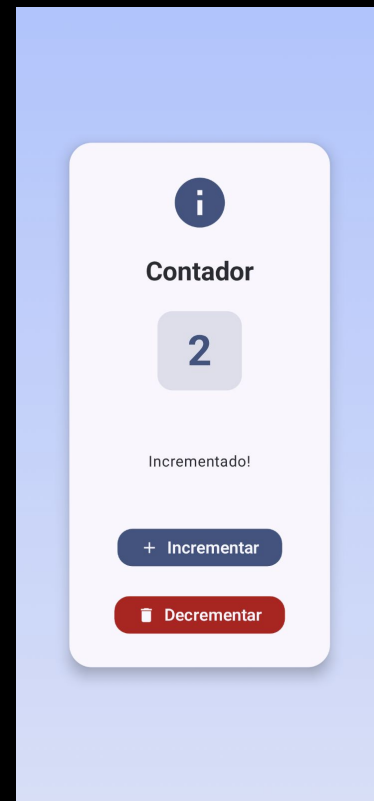
Adicionando Botão

Estrutura do Layout

Implementações

Aqui, fazemos os ajustes no botão decrementar.

```
135     shape = RoundedCornerShape( size = 16.dp),
136     modifier = Modifier.padding(horizontal = 16.dp)
137   ) {
138     Icon(
139       imageVector = Icons.Default.Add,
140       contentDescription = "Incrementar",
141       modifier = Modifier.size( size = 20.dp)
142     )
143     Text(
144       text = " Incrementar",
145       fontSize = 18.sp,
146       fontWeight = FontWeight.SemiBold
147     )
148   }
149
150   // Botão Decrementar
151   Button(
152     // onClick = {
153     //   if (contador > 0) {
154     //     contador--
155     //   }
156     // },
157     onClick = { viewModel.decrementar() },
158     colors = ButtonDefaults.buttonColors(
159       containerColor = MaterialTheme.colorScheme.error
160     ),
161     shape = RoundedCornerShape( size = 16.dp),
162     modifier = Modifier.padding(horizontal = 16.dp)
163   ) {
164     Icon(
165       imageVector = Icons.Default.Delete,
166       contentDescription = "Decrementar",
167       modifier = Modifier.size( size = 20.dp)
168     )
169     Text(
170       text = " Decrementar",
171       fontSize = 18.sp,
172       fontWeight = FontWeight.SemiBold
173     )
174   }
175 }
176
177 }
178 }
```





Fernando Sambinelli

Professor e Doutor em Tecnologia
sambinelli@ifsp.edu.br



INSTITUTO FEDERAL
São Paulo

